Chapter 6: Bessel Functions

Special Functions of Mathematical Physics: A Tourist's Guidebook

after K. A. Novak
Compiled by Jake W. Liu

June 12, 2026

Outline

Where we are going

Bessel's equation,

$$x^2 J_m'' + x J_m' + (x^2 - m^2) J_m = 0,$$

arose when we separated variables on a circular drum head. The same equation appears in separation of variables for the Laplace and Helmholtz equations in spherical and cylindrical coordinates, and in electromagnetic waves, heat conduction, vibration, and Schrödinger's equation.

- We package all the Bessel functions $J_m(x)$ into a single **generating function** and read identities straight off it.
- Differentiating the generating function gives **recursion formulae**.
- A clever substitution turns it into an **integral representation**, and a contour version gives the **Schläfli integral**.
- Steepest descent on that contour yields the **large-x asymptotics**.
- A second solution—the **Neumann function Y_m**—and the **Hankel transform** complete the picture.

Table of contents

What a generating function is

We call $g(x, t)$ a *generating function* of the functions $P_m(x)$ if its Laurent expansion in t has those functions as coefficients:

$$g(x, t) = \sum_{m=-\infty}^{\infty} P_m(x)\, t^m. \tag{6.1.1}$$

The whole infinite family $\{P_m\}$ is then stored in one object: any operation we can do to g (differentiate, substitute a value of t, expand) turns into a statement about the P_m once we match powers of t.
The generating function for the Bessel functions can be motivated by expanding a plane wave in circular modes, but we do not need to assume that result. We can define the coefficients of the Laurent expansion below and then read them off directly; the coefficient calculation on the next slide reproduces exactly the Frobenius series for J_m.

What a generating function is (cont.)

The generating function for the Bessel functions is

$$g(x,t) = e^{\frac{1}{2}x\left(t - t^{-1}\right)} = \sum_{m=-\infty}^{\infty} J_m(x)\, t^m. \tag{6.1.2}$$

Here $J_m(x)$ is the *Bessel function of the first kind* of order m: the solution of Bessel's equation that stays finite at the origin. Equation (6.1.2) is the single identity from which this chapter is built; everything else follows by manipulating it.

Reading off the series solution

To extract the coefficient of t^m, split the exponential into two ordinary exponentials and expand each with $e^u = \sum_{n\geq 0} u^n/n!$. For the first factor $u = \frac{x}{2}t$, so $u^j = (\frac{x}{2})^j t^j$; for the second factor $u = -\frac{x}{2t}$, so $u^k = (-1)^k(\frac{x}{2})^k t^{-k}$ (the $(-1)^k$ comes from the minus sign in the exponent):

$$g(x,t) = e^{xt/2}\, e^{-x/(2t)} = \left(\sum_{j=0}^{\infty} \frac{1}{j!}\left(\tfrac{x}{2}\right)^j t^j\right)\left(\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}\left(\tfrac{x}{2}\right)^k t^{-k}\right).$$

Multiplying the two series, the power of t in a product term is $j-k$. Collecting all terms with $j-k=m$,

$$g(x,t) = \sum_{m=-\infty}^{\infty}\left(\sum_{\substack{j-k=m\\ j,k\geq 0}} \frac{(-1)^k}{j!\,k!}\left(\tfrac{x}{2}\right)^{j+k}\right) t^m.$$

Reading off the series solution (cont.)

First take $m = 0, 1, 2, \ldots$. Set $j = k + m$ to remove the constraint; then $j + k = 2k + m$ and the inner sum runs over $k \geq 0$:

$$[t^m]\, g(x,t) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(m+k)!\, k!} \left(\tfrac{x}{2}\right)^{2k+m}.$$

Reading off the series solution (cont.)

Matching the coefficient of t^m gives the *series solution*

$$J_m(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,(m+k)!} \left(\tfrac{x}{2}\right)^{2k+m}. \tag{6.1.3}$$

This is exactly the Frobenius series found earlier, which confirms that the stated generating function (6.1.2) is the right one. The negative integer orders are handled next by comparing J_{-m} with J_m.
For *general* (non-integer) order m the factorial $(m+k)!$ is replaced by the gamma function $\Gamma(m+k+1)$, which interpolates the factorial:

$$J_m(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(m+k+1)} \left(\tfrac{x}{2}\right)^{2k+m}. \tag{6.1.4}$$

Negative orders: $J_{-m} = (-1)^m J_m$

Replacing m by $-m$ in (6.1.3) gives

$$J_{-m}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,(-m+k)!} \left(\tfrac{x}{2}\right)^{-m+2k}.$$

For $k = 0, 1, \ldots, m-1$, read the factorial as $\Gamma(-m+k+1)$. Its argument is a nonpositive integer, so Γ has a pole and $1/\Gamma(-m+k+1) = 0$. Those terms vanish, and the sum effectively starts at $k = m$:

$$J_{-m}(x) = \sum_{k=m}^{\infty} \frac{(-1)^k}{k!\,(-m+k)!} \left(\tfrac{x}{2}\right)^{-m+2k}.$$

Shift the index, $k \to k + m$, so the new k again starts at 0:

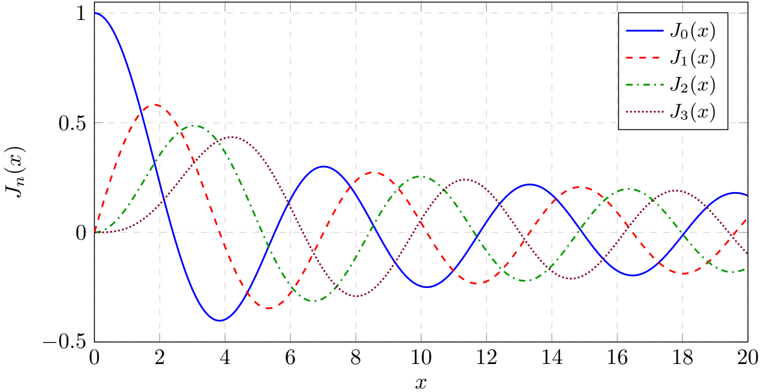
$$J_{-m}(x) = \sum_{k=0}^{\infty} \frac{(-1)^{k+m}}{(k+m)!\,k!} \left(\tfrac{x}{2}\right)^{m+2k} = (-1)^m \sum_{k=0}^{\infty} \frac{(-1)^k}{(k+m)!\,k!} \left(\tfrac{x}{2}\right)^{m+2k}.$$

The remaining sum is precisely $J_m(x)$ from (6.1.3), so

$$J_{-m}(x) = (-1)^m J_m(x), \qquad m \in \mathbb{Z}. \tag{6.1.5}$$

Negative integer orders are not new functions; they are just signed copies.

Bessel functions of the first kind



The regular solution J_m stays finite at the origin. Higher order forces the leading behavior $J_m(x) \sim (x/2)^m/\Gamma(m+1)$, while all fixed orders eventually oscillate with decaying envelope.

Example: a discrete Fourier series in disguise

Generating functions let us read off useful identities quickly. Take $t = e^{i\theta}$ in (6.1.2). Then

$$t - t^{-1} = e^{i\theta} - e^{-i\theta} = 2i \sin\theta,$$

so the exponent becomes $\frac{1}{2}x\,(2i\sin\theta) = ix\sin\theta$ and

$$e^{\frac{1}{2}x\left(t-t^{-1}\right)} = e^{ix\sin\theta}. \tag{6.1.6}$$

The right side of (6.1.2) with $t = e^{i\theta}$ is $\sum_m J_m(x)\,e^{im\theta}$, so

$$\sum_{m=-\infty}^{\infty} J_m(x)\,e^{im\theta} = e^{ix\sin\theta}. \tag{6.1.7}$$

Read as a function of θ, the right side is periodic and the left side is its Fourier series; the Bessel functions $J_m(x)$ are the *discrete Fourier coefficients* of $e^{ix\sin\theta}$. We exploit this in Section 6.3.

Table of contents

Differentiating in t: the three-term recursion

Differentiate the generating function (6.1.2) with respect to t. On the closed form,

$$\frac{\partial g}{\partial t} = \frac{x}{2}\left(1 + t^{-2}\right) e^{\frac{1}{2}x(t-t^{-1})} = \frac{x}{2}\left(1 + t^{-2}\right) g(x,t),$$

where the bracket is $\frac{\partial}{\partial t}\frac{1}{2}x(t - t^{-1}) = \frac{x}{2}(1 + t^{-2})$. Substitute the series for g:

$$\frac{\partial g}{\partial t} = \frac{x}{2}\left(1 + t^{-2}\right) \sum_{m=-\infty}^{\infty} J_m(x)\, t^m = \frac{x}{2}\left(\sum_m J_m(x)\, t^m + \sum_m J_m(x)\, t^{m-2}\right).$$

In the second sum shift $m \to m + 2$ so both run over t^m:

$$\frac{\partial g}{\partial t} = \frac{x}{2} \sum_{m=-\infty}^{\infty} \left(J_m(x) + J_{m+2}(x)\right) t^m. \tag{6.2.1}$$

Differentiating in t: the three-term recursion (cont.)

Now differentiate the *series* form $\sum_m J_m(x)\, t^m$ instead. Term by term,

$$\frac{\partial g}{\partial t} = \sum_{m=-\infty}^{\infty} m\, J_m(x)\, t^{m-1} = \sum_{m=-\infty}^{\infty} (m+1)\, J_{m+1}(x)\, t^m,$$

where the last step shifts $m \to m+1$. The two expressions for $\partial g/\partial t$ must agree as Laurent series, so their coefficients of t^m match:

$$\frac{x}{2}\big(J_m(x) + J_{m+2}(x)\big) = (m+1)\, J_{m+1}(x).$$

Replacing $m+1 \to m$ (so $m \to m-1$ and $m+2 \to m+1$) gives the cleaner form

$$J_{m+1}(x) + J_{m-1}(x) = \frac{2m}{x}\, J_m(x). \tag{6.2.2}$$

This three-term recurrence is used for $x \neq 0$. Endpoint values at $x = 0$, when they are meaningful, are read by limits from the series. Away from the origin it climbs the orders: knowing J_m and J_{m-1} gives J_{m+1}.

Differentiating in x: the derivative recursion

Now differentiate the generating function (6.1.2) with respect to x. On the closed form the exponent has x-derivative $\frac{1}{2}(t - t^{-1})$, so

$$\frac{\partial g}{\partial x} = \tfrac{1}{2}\left(t - t^{-1}\right) e^{\frac{1}{2}x(t-t^{-1})} = \tfrac{1}{2}\left(t - t^{-1}\right) g(x,t).$$

Substitute the series and split:

$$\frac{\partial g}{\partial x} = \tfrac{1}{2}\left(t - t^{-1}\right) \sum_m J_m(x)\, t^m = \tfrac{1}{2} \sum_m J_m(x)\, t^{m+1} - \tfrac{1}{2} \sum_m J_m(x)\, t^{m-1}.$$

Shift $m \to m-1$ in the first sum and $m \to m+1$ in the second so both carry t^m:

$$\frac{\partial g}{\partial x} = \tfrac{1}{2} \sum_{m=-\infty}^{\infty} \left(J_{m-1}(x) - J_{m+1}(x)\right) t^m. \tag{6.2.3}$$

Differentiating in x: the derivative recursion (cont.)

The series form differentiates term by term to $\partial g/\partial x = \sum_m J'_m(x)\, t^m$. Equating coefficients of t^m with (6.2.3),

$$J'_m(x) = \tfrac{1}{2}\, J_{m-1}(x) - \tfrac{1}{2}\, J_{m+1}(x), \quad \text{equivalently} \quad 2J'_m(x) = J_{m-1}(x) - J_{m+1}(x). \quad (6.2.4)$$

The two recursions (6.2.2) and (6.2.4) together let us compute every derivative and every order starting from J_0 and J_1 alone. To see this, use (6.2.2) for $x \neq 0$ in the form $J_{m\mp1}(x) = \frac{2m}{x} J_m(x) - J_{m\pm1}(x)$ to eliminate one neighbour from (6.2.4). Eliminating J_{m+1} (replace it by $\frac{2m}{x} J_m - J_{m-1}$) in $2J'_m = J_{m-1} - J_{m+1}$ gives $2J'_m = 2J_{m-1} - \frac{2m}{x} J_m$; eliminating J_{m-1} instead gives $2J'_m = \frac{2m}{x} J_m - 2J_{m+1}$. Dividing each by 2,

$$J'_m(x) = J_{m-1}(x) - \frac{m}{x} J_m(x), \qquad J'_m(x) = \frac{m}{x} J_m(x) - J_{m+1}(x),$$

and at $m = 0$ the second reads $J'_0(x) = -J_1(x)$.

Table of contents

From the generating function to a Fourier sum

Start again from

$$e^{\frac{1}{2}x(t-t^{-1})} = \sum_{m=-\infty}^{\infty} J_m(x)\, t^m, \tag{6.3.1}$$

and set $t = e^{i\theta}$ (as in the Section 6.1 example), giving $e^{ix\sin\theta} = \sum_m J_m(x)\, e^{im\theta}$. Pull out the $m = 0$ term and pair each $m > 0$ with its $-m$ partner. Using $J_{-m} = (-1)^m J_m$ from (6.1.5),

$$\begin{aligned} e^{ix\sin\theta} &= J_0(x) + \sum_{m=1}^{\infty} \Big(J_m(x)\, e^{im\theta} + J_{-m}(x)\, e^{-im\theta} \Big) \\ &= J_0(x) + \sum_{m=1}^{\infty} J_m(x) \Big(e^{im\theta} + (-1)^m\, e^{-im\theta} \Big). \end{aligned}$$

Split the surviving sum by the parity of m, writing $m = 2\ell$ for even and $m = 2\ell + 1$ for odd to avoid clashing with the running index. For even $m = 2\ell$ the factor $(-1)^m = +1$, so $e^{im\theta} + (-1)^m e^{-im\theta} = e^{i2\ell\theta} + e^{-i2\ell\theta} = 2\cos(2\ell\theta)$; for odd $m = 2\ell + 1$ the factor $(-1)^m = -1$, so $e^{im\theta} + (-1)^m e^{-im\theta} = e^{i(2\ell+1)\theta} - e^{-i(2\ell+1)\theta} = 2i\sin\big((2\ell+1)\theta\big)$.

From the generating function to a Fourier sum (cont.)

The original sum runs over $m \geq 1$. The even values $m = 2, 4, \dots$ correspond to $\ell = 1, 2, \dots$, and the odd values $m = 1, 3, \dots$ correspond to $\ell = 0, 1, \dots$, which fixes the two lower limits:

$$
\begin{aligned}
e^{ix\sin\theta} = J_0(x) + 2\sum_{\ell=1}^{\infty} J_{2\ell}(x)\cos(2\ell\theta) \\
+ 2i\sum_{\ell=0}^{\infty} J_{2\ell+1}(x)\sin\big((2\ell+1)\theta\big).
\end{aligned}
\tag{6.3.2}
$$

On the other hand, the left side is literally

$$
e^{ix\sin\theta} = \cos(x\sin\theta) + i\sin(x\sin\theta).
$$

Equate real and imaginary parts. The real part is J_0 plus the even-order cosine sum; the imaginary part is the odd-order sine sum. Renaming the dummy index back to m (and writing the odd sum with $m = \ell + 1$, so $2\ell + 1 = 2m - 1$ and m starts at 1),

$$
\cos(x\sin\theta) = J_0(x) + 2\sum_{m=1}^{\infty} J_{2m}(x)\cos(2m\theta), \tag{6.3.3}
$$

From the generating function to a Fourier sum (cont.)

$$\sin(x \sin\theta) = 2 \sum_{m=1}^{\infty} J_{2m-1}(x) \sin\big((2m-1)\theta\big). \tag{6.3.4}$$

Projecting out J_m by orthogonality

The expansions (6.3.3)–(6.3.4) are Fourier cosine/sine series, so we recover a single J_m by integrating against the matching harmonic. On $[0, \pi]$, distinct cosine modes are orthogonal; the cosine norm is π for the zero mode and $\pi/2$ for modes $n \geq 1$. For sine modes,

$$\int_0^\pi \sin n\theta \, \sin m\theta \, d\theta = \tfrac{\pi}{2}\, \delta_{nm} \qquad (n, m \geq 1).$$

Multiply (6.3.3) by $\cos(m\theta)$ and integrate. For $m = 0$, the J_0 term integrates to $\pi J_0(x)$. For positive even m, the matching term carries the coefficient 2 from (6.3.3) times the orthogonality value $\pi/2$, again giving $\pi J_m(x)$; for odd m no term matches and the integral is 0. The same applies to (6.3.4) with $\sin(m\theta)$, with the roles of the parities swapped. The result is, with the parity condition stated on m (not on any other index),

$$\int_0^\pi \cos(x \sin\theta)\cos(m\theta)\, d\theta = \begin{cases} \pi J_m(x), & m \text{ even}, \\ 0, & m \text{ odd}, \end{cases}$$

$$\int_0^\pi \sin(x \sin\theta)\sin(m\theta)\, d\theta = \begin{cases} 0, & m \text{ even}, \\ \pi J_m(x), & m \text{ odd}. \end{cases}$$

Projecting out J_m by orthogonality (cont.)

For any given integer $m \geq 0$, exactly one of the two integrals is nonzero and equals $\pi J_m(x)$, so we may simply *add* them: the irrelevant case contributes 0, and the sum is $\pi J_m(x)$ either way. Dividing by π,

$$J_m(x) = \frac{1}{\pi} \int_0^{\pi} \cos\Big(m\theta - x \sin\theta\Big)\, d\theta, \tag{6.3.5}$$

using the cosine difference identity $\cos A \cos B + \sin A \sin B = \cos(A - B)$ with $A = x \sin\theta$, $B = m\theta$.

For the special case $m = 0$ the cosine integral above already gives J_0, and dropping the $m\theta$ term leaves

$$J_0(x) = \frac{1}{\pi} \int_0^{\pi} \cos(x \sin\theta)\, d\theta. \tag{6.3.6}$$

Table of contents

A contour integral for J_n

A representation of a function by a contour integral is called a *Schläfli integral*. Divide the generating function (6.1.2) by t^{n+1} for a fixed integer n and integrate around a positively oriented closed contour that winds once around the origin:

$$\oint e^{\frac{1}{2}x(t-t^{-1})}\, t^{-n-1}\, dt = \oint \sum_{m=-\infty}^{\infty} J_m(x)\, t^{m-n-1}\, dt = \sum_{m=-\infty}^{\infty} J_m(x) \oint t^{m-n-1}\, dt.$$

The basic residue fact $\oint t^{k}\, dt = 2\pi i\, \delta_{k,-1}$ (nonzero only when the exponent is -1, i.e. $m = n$) collapses the sum:

$$\oint e^{\frac{1}{2}x(t-t^{-1})}\, t^{-n-1}\, dt = \sum_{m} J_m(x)\, 2\pi i\, \delta_{mn} = 2\pi i\, J_n(x).$$

A contour integral for J_n (cont.)

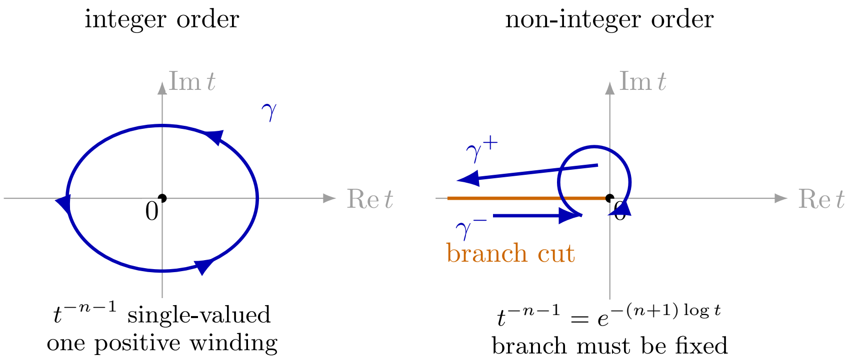
Solving for J_n gives the *Schläfli integral*:

$$J_n(x) = \frac{1}{2\pi i} \oint e^{\frac{1}{2}x(t-t^{-1})}\, t^{-n-1}\, dt. \tag{6.4.1}$$

The integrand is single-valued for integer n, so any positively oriented contour winding once around the origin works.

For *non-integer n* the factor t^{-n-1} has a branch cut, and the contour must be deformed to avoid it: instead of a closed loop around the origin we use a path that comes in from $-\infty$ below the cut, wraps the origin, and returns to $-\infty$ above the cut. That Hankel-style version needs a stated branch and normalization; the closed-loop formula (6.4.1) is the integer-order formula used below. This contour form is the launching point for the large-x asymptotics.

Contours in the Schläfli representation



For integer n, t^{-n-1} is single-valued and Cauchy's coefficient formula only needs a closed contour with winding number one around 0. For non-integer n, the same factor is $e^{-(n+1)\log t}$, so the branch of $\log t$ must be chosen and the contour must avoid the cut. This is the contour issue hidden behind the phrase "continue to non-integer order."

Table of contents

Small x: the leading power

When x is small, $x \ll 1$, the series (6.1.3) is dominated by its first term $k = 0$, because every higher term carries an extra factor $(x/2)^2$. Keeping only $k = 0$,

$$J_m(x) \approx \frac{1}{m!}\left(\frac{x}{2}\right)^m = \frac{1}{\Gamma(m+1)}\left(\frac{x}{2}\right)^m, \qquad x \ll 1. \tag{6.5.1}$$

So near the origin J_m behaves like a pure power x^m: $J_0 \to 1$, while $J_m \to 0$ for $m \geq 1$, vanishing faster the higher the order. This is the regular behavior we demanded of the drum solution at its center.

Large x: setting up steepest descent

For fixed integer n and $x \to +\infty$, we apply the method of steepest descent to the Schläfli integral. Recall the general estimate: for $s \gg 1$,

$$\int_C g(z)\, e^{sf(z)}\, dz \approx \frac{\sqrt{2\pi}\, g(z_0)\, e^{sf(z_0)}\, i\, e^{-i\vartheta/2}}{\sqrt{|sf''(z_0)|}}, \qquad \vartheta = \arg f''(z_0), \tag{6.5.2}$$

evaluated at a saddle point z_0 where $f'(z_0) = 0$. Writing the Schläfli integrand (6.4.1) in the form $g(t)\, e^{xf(t)}$ with $s = x$, we identify

$$f(t) = \tfrac{1}{2}\left(t - t^{-1}\right), \qquad g(t) = t^{-(n+1)}. \tag{6.5.3}$$

Large x: setting up steepest descent (cont.)

The saddle points come from $f'(t) = 0$:

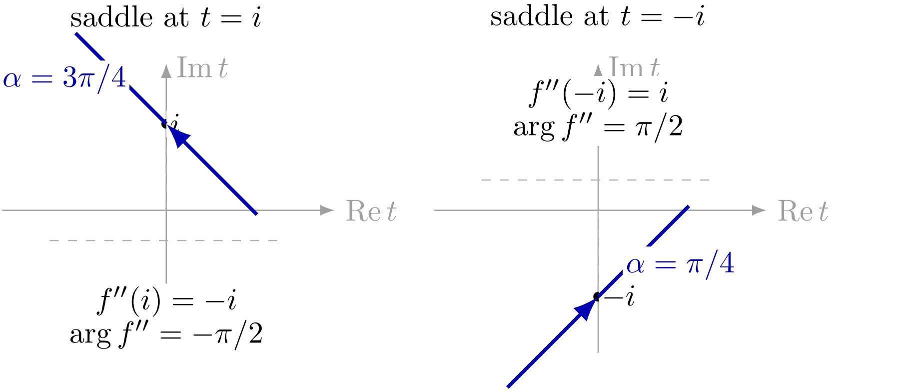
$$f'(t) = \tfrac{1}{2}\left(1 + t^{-2}\right) = 0 \implies t^{-2} = -1 \implies t = \pm i.$$

There are *two* saddle points, $t = i$ and $t = -i$; both contribute. With $f''(t) = -t^{-3}$ we evaluate at each:

$$\begin{array}{ll} f(i) = \tfrac{1}{2}(i + i) = i & f(-i) = \tfrac{1}{2}(-i - i) = -i \\ f''(i) = -i & f''(-i) = i \\ g(i) = e^{-(n+1)i\pi/2} & g(-i) = e^{(n+1)i\pi/2} \\ \vartheta = \arg f''(i) = -\pi/2 & \vartheta = \arg f''(-i) = \pi/2 \end{array}$$

(Here $i = e^{i\pi/2}$ so $i^{n+1} = e^{(n+1)i\pi/2}$, and $f''(i) = -(i)^{-3} = -i$.) The steepest-descent direction is $\alpha = \pi/2 - \vartheta/2$, so the contour is deformed to cross $t = i$ at angle $3\pi/4$ and $t = -i$ at angle $\pi/4$. The factor $ie^{-i\vartheta/2}$ in (6.5.2) records this chosen orientation through each saddle; choosing the opposite orientation would flip the local sign, which the full contour orientation fixes.

Steepest-descent paths through the two saddles



The local descent direction is

$$\alpha = \frac{\pi}{2} - \frac{1}{2} \arg f''(t_0).$$

Thus $f''(i) = -i$ gives $\alpha = 3\pi/4$, while $f''(-i) = i$ gives $\alpha = \pi/4$. These two local contributions must both be kept; dropping either saddle would lose the final cosine.

Large x: combining the two saddles

Apply the master estimate (6.5.2) at each saddle. Since $|sf''| = x$ at both points, $\sqrt{|sf''|} = \sqrt{x}$. The master factor $e^{-i\vartheta/2}$ uses $\vartheta = \arg f''$: at $t = i$, $\vartheta = -\pi/2$ gives $e^{-i\vartheta/2} = e^{i\pi/4}$; at $t = -i$, $\vartheta = \pi/2$ gives $e^{-i\vartheta/2} = e^{-i\pi/4}$. Substituting the table values,

$$I_i = \frac{\sqrt{2\pi}}{\sqrt{x}}\, e^{-(n+1)i\pi/2}\, e^{ix}\, i\, e^{i\pi/4}.$$

Collect the three phases into one exponent. Pulling out the constant $e^{-i\pi/2}$ from $e^{-(n+1)i\pi/2} = e^{-in\pi/2}\, e^{-i\pi/2}$ and absorbing $e^{-i\pi/2}\, e^{i\pi/4} = e^{-i\pi/4}$, the combined phase is $-in\pi/2 + ix - i\pi/4 = -i(n\pi/2 - x + \pi/4)$, so

$$I_i = \frac{\sqrt{2\pi}}{\sqrt{x}}\, i\, e^{-i(n\pi/2 - x + \pi/4)}.$$

The saddle at $t = -i$ is the complex conjugate combination $(e^{(n+1)i\pi/2}\, e^{-ix}\, e^{-i\pi/4})$, giving the same modulus with the opposite sign in the exponent:

$$I_{-i} = \frac{\sqrt{2\pi}}{\sqrt{x}}\, e^{(n+1)i\pi/2}\, e^{-ix}\, i\, e^{-i\pi/4} = \frac{\sqrt{2\pi}}{\sqrt{x}}\, i\, e^{i(n\pi/2 - x + \pi/4)}.$$

Large x: combining the two saddles (cont.)

The Schläfli prefactor is $1/(2\pi i)$, so the two saddles combine as

$$J_n(x) \approx \frac{1}{2\pi i}\big(I_i + I_{-i}\big). \tag{6.5.4}$$

Factor out the common $\dfrac{1}{2\pi i} \cdot \dfrac{\sqrt{2\pi}}{\sqrt{x}}\, i = \dfrac{1}{\sqrt{2\pi x}}$ and use $e^{-i\phi} + e^{i\phi} = 2\cos\phi$ with $\phi = n\pi/2 - x + \pi/4$:

$$\begin{aligned} J_n(x) &\approx \frac{1}{\sqrt{2\pi x}}\Big(e^{-i(n\pi/2 - x + \pi/4)} + e^{i(n\pi/2 - x + \pi/4)}\Big) \\ &= \frac{2}{\sqrt{2\pi x}}\cos\Big(n\pi/2 - x + \pi/4\Big) = \sqrt{\frac{2}{\pi x}}\cos\Big(x - \frac{n\pi}{2} - \frac{\pi}{4}\Big), \end{aligned}$$

since cosine is even. The large-x asymptotic is therefore

$$J_n(x) \approx \sqrt{\frac{2}{\pi x}}\cos\Big(x - \frac{n\pi}{2} - \frac{\pi}{4}\Big), \qquad x \to +\infty, \quad n \text{ fixed}. \tag{6.5.5}$$

Far from the origin J_n is a decaying cosine: amplitude $\sim x^{-1/2}$, phase shifted by $n\pi/2 + \pi/4$.

Table of contents

The second solution Y_m

For integer order m, the method of Frobenius produces only *one* independent solution of Bessel's equation, namely J_m. A second-order ODE has two independent solutions, so a second one must exist. Reduction of order supplies it on any interval $x > 0$ where J_m has no zero. Starting from J_m,

$$y(x) = A\, J_m(x) + B\, J_m(x) \int_{x_*}^{x} \frac{1}{x'\, J_m^2(x')}\, dx' = A'\, J_m(x) + B'\, Y_m(x), \tag{6.6.1}$$

where Y_m is the *Bessel function of the second kind*, also called the *Neumann function*. Here $x_* > 0$ is any convenient base point in that zero-free interval. For integer m, Y_m is singular at the origin: Y_0 has a logarithmic singularity, while Y_m for $m \geq 1$ has an algebraic x^{-m} singularity. It is therefore *unphysical* for the circular drum, whose displacement must be finite at the center. That is why only J_m appeared in the drum solution.

The second solution Y_m (cont.)

For *non-integer* m the Neumann function has a clean closed form built from J_m and J_{-m}, which are then independent:

$$Y_m(x) = \frac{J_m(x)\cos(m\pi) - J_{-m}(x)}{\sin(m\pi)}. \tag{6.6.2}$$

Being a combination of solutions, Y_m is itself a solution. When m is an integer this formula is indeterminate: using $J_{-m} = (-1)^m J_m$ and $\cos(m\pi) = (-1)^m$,

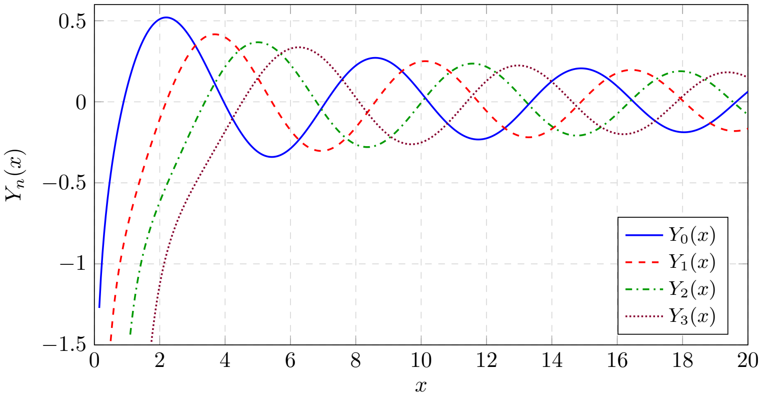
$$Y_m(x) = \frac{J_m(x)(-1)^m - (-1)^m J_m(x)}{\sin(m\pi)} = \frac{0}{0}.$$

The integer case is recovered by l'Hôpital's rule. Let ν be the order and take $\nu \to m$:

$$Y_m(x) = \frac{(-1)^m \, \partial_\nu J_\nu(x)|_{\nu=m} - \partial_\nu J_{-\nu}(x)|_{\nu=m}}{\pi(-1)^m}.$$

The denominator derivative is $\frac{d}{d\nu}\sin(\pi\nu)|_{\nu=m} = \pi(-1)^m$. The order derivatives in the numerator differentiate factors such as $(x/2)^{\pm\nu}$ in the Bessel series, and $\partial_\nu(x/2)^\nu = (x/2)^\nu \log(x/2)$; this is where the logarithmic singularity of Y_m enters.

Bessel functions of the second kind



The Neumann functions Y_m are independent of J_m but singular at $x = 0$. Away from the origin, J_m and Y_m form the decaying cosine/sine pair behind the Hankel functions.

Asymptotics of J_m and Y_m

For fixed $m \geq 0$ and positive real x, the small- and large-argument behaviors are:

$$J_m(x) \approx \frac{1}{\Gamma(m+1)}\left(\tfrac{x}{2}\right)^m, \qquad x \to 0^+, \tag{6.6.3}$$

$$J_m(x) \approx \sqrt{\frac{2}{\pi x}}\cos\!\left(x - \frac{m\pi}{2} - \frac{\pi}{4}\right), \qquad x \to +\infty, \tag{6.6.4}$$

and for the Neumann functions

$$Y_m(x) \approx -\frac{\Gamma(m)}{\pi}\left(\tfrac{2}{x}\right)^m, \qquad x \to 0^+ \ \ (m > 0), \tag{6.6.5}$$

$$Y_m(x) \approx \sqrt{\frac{2}{\pi x}}\sin\!\left(x - \frac{m\pi}{2} - \frac{\pi}{4}\right), \qquad x \to +\infty. \tag{6.6.6}$$

For $m = 0$, $Y_0(x)$ instead has logarithmic behavior, $Y_0(x) \sim (2/\pi)\log x$ up to an additive constant. Near the origin Y_m is singular; far from the origin J_m and Y_m are a cos/sin pair of the same decaying amplitude $\sqrt{2/\pi x}$ and the same phase $x - m\pi/2 - \pi/4$: they behave as complementary decaying trigonometric functions, just like cos and sin underlie $e^{\pm i\theta}$.

Table of contents

Hankel functions: the "third kind"

Just as the exponential is built from trigonometric functions, $e^{\pm i\theta} = \cos\theta \pm i\sin\theta$, we combine the two Bessel solutions into *Hankel functions* (Bessel functions of the third kind):

$$H_m^{(1)}(x) = J_m(x) + i\,Y_m(x), \tag{6.7.1}$$

$$H_m^{(2)}(x) = J_m(x) - i\,Y_m(x). \tag{6.7.2}$$

By the large-x asymptotics (6.6.4) and (6.6.6), far from the origin

$$H_m^{(1)}(x) \approx \sqrt{\frac{2}{\pi x}}\, e^{\,i(x - m\pi/2 - \pi/4)},$$

which is an outgoing radial wave if the time factor is $e^{-i\omega t}$. With that same time convention, $H_m^{(2)}$ is incoming. Because Y_m is singular at the origin, so is each Hankel function; one should not use a Hankel approximation in place of a J_m approximation *near* the origin in numerical work.

Why a Hankel transform should exist

The analogy is the same one used for the string, but with the radial eigenfunctions changed. On a finite clamped string the spatial modes are sines:

$$u(t,x) = \sum_{n=1}^{\infty} a_n \cos(cn\pi t)\sin(n\pi x), \qquad a_m = 2\int_0^1 f(x)\sin(m\pi x)\,dx.$$

On the whole real line the discrete sine coefficients are replaced by Fourier coefficients, with one common normalization

$$\widehat{f}(k) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)e^{-ikx}\,dx, \qquad f(x) = \int_{-\infty}^{\infty} \widehat{f}(k)e^{ikx}\,dk.$$

Why a Hankel transform should exist (cont.)

For a circular drum, the radial equation has Bessel eigenfunctions instead of sines. For the radially symmetric angular mode $m = 0$ on a unit disk,

$$u(t,r) = \sum_{j=1}^{\infty} a_j \cos(c\alpha_{0,j} t) J_0(\alpha_{0,j} r),$$

where $\alpha_{0,j}$ is the j-th positive zero of J_0. On the infinite radial half-line, the discrete zeros become a continuous radial frequency k, and the coefficient integral is the Hankel transform

$$\mathcal{H}_m[f](k) = \int_0^{\infty} f(r) J_m(kr)\, r\, dr.$$

The extra factor r is not optional: it is the polar area weight, just as the finite disk inner product uses the weight s.

Eigenfunctions and the Sturm–Liouville form

The finite-disk version follows from Sturm–Liouville theory. For the string equation the sinusoidal modes are eigenfunctions; with clamped endpoints on $[0, 1]$, the allowed modes are $\sin n\pi x$:

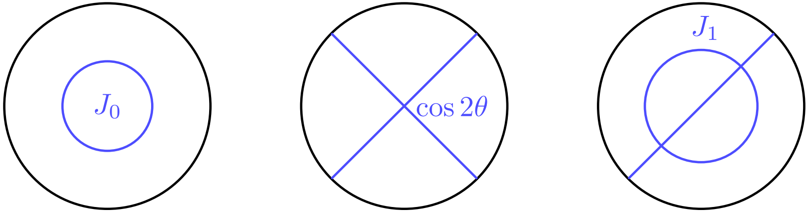
$$u(t,x) = \sum_{n=1}^{\infty} a_n \cos(cn\pi t)\,\sin(n\pi x), \qquad a_m = 2\int_0^1 f(x)\,\sin(m\pi x)\,dx,$$

while J_0 gives the radially symmetric modes for a circular drum,

$$u(t,r) = \sum_{j=1}^{\infty} a_j \cos(c\alpha_{0j} t)\, J_0(\alpha_{0j} r),$$

where α_{0j} is the j-th positive zero of J_0.

Eigenfunctions and the Sturm–Liouville form (cont.)



$m = 0,\ j = 2$ $\quad$ $m = 2,\ j = 1$ $\quad$ $m = 1,\ j = 2$

Because the drum is clamped on its rim, we stretch each Bessel function so that one of its zeros lands on the boundary $x = 1$. Substituting $x = \alpha s$ into Bessel's equation,

$$s^2 \frac{d^2 y}{ds^2} + s\frac{dy}{ds} + (\alpha^2 s^2 - m^2)\, y = 0. \tag{6.7.3}$$

Eigenfunctions and the Sturm–Liouville form (cont.)

Dividing by s and grouping the first two terms as a single derivative, the *Sturm–Liouville form* is

$$\frac{d}{ds}\left(s\,\frac{dy}{ds}\right) - \frac{m^2}{s}\,y = -\alpha^2 s\,y. \tag{6.7.4}$$

(The first term expands as $\frac{\mathrm{d}}{\mathrm{d}s}(sy') = sy'' + y'$, which matches $y'' + \frac{1}{s}y'$ after multiplying (6.7.3) by $1/s$; the sign on the right is $-\alpha^2 s\,y$.) In the Sturm–Liouville convention $(py')' + qy = -\lambda wy$, this identifies $\lambda = \alpha^2$ with weight $w = s$. The Bessel functions are therefore orthogonal under the weighted inner product

$$(u, v) = \int_0^1 u\,v\,s\,ds. \tag{6.7.5}$$

At $s = 0$ the boundary term is still harmless for the regular modes: $J_m(\alpha s) = O(s^m)$ and $J_m'(\alpha s) = O(s^{m-1})$ (with $J_0'(\alpha s) = O(s)$), hence $s(y_i y_j' - y_j y_i') \to 0$. The singular Y_m modes are not allowed at the drum center.

Fourier–Bessel series and the Hankel transform

Let $\alpha_{m,j}$ denote the positive zeros of J_m (so $J_m(\alpha_{m,j}) = 0$). Orthogonality with the weight s takes the explicit form

$$\int_0^1 J_m(\alpha_{m,j}x)\, J_m(\alpha_{m,k}x)\, x\, dx = \tfrac{1}{2}\, J_{m+1}^2(\alpha_{m,j})\, \delta_{jk}. \tag{6.7.6}$$

Just as a function on an interval expands in sinusoids, the radial part of a fixed angular mode on the disk expands in a *Fourier–Bessel series* of order m,

$$f(x) = \sum_{j=1}^{\infty} a_j\, J_m(\alpha_{m,j}x), \tag{6.7.7}$$

with coefficients read off by projecting onto the orthogonal mode:

$$a_j = \frac{2}{J_{m+1}^2(\alpha_{m,j})} \int_0^1 f(r)\, J_m(\alpha_{m,j}r)\, r\, dr. \tag{6.7.8}$$

The full disk expansion also includes the angular factors $\cos(m\theta)$ and $\sin(m\theta)$ (or $e^{im\theta}$). On the infinite half-line the corresponding continuous tool is the Hankel transform $\mathcal{H}_m[f](k) = \int_0^\infty f(r) J_m(kr)\, r\, dr$.

Example: Fourier transform is the order-zero Hankel transform

The two-dimensional Fourier transform is

$$F(k_x, k_y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, e^{-ik_x x} e^{-ik_y y}\, dx\, dy. \tag{6.7.9}$$

Pass to polar coordinates. With $\mathbf{k} \cdot \mathbf{x} = kr \sin\theta$ (a suitable orientation of axes) and area element $r\, dr\, d\theta$,

$$F(k_x, k_y) = \frac{1}{2\pi} \int_0^{2\pi} \int_0^{\infty} f(r, \theta)\, e^{-ikr \sin\theta}\, r\, dr\, d\theta. \tag{6.7.10}$$

Note the exponent is $-ikr \sin\theta$: the full dot product carries the radius r.

Example: Fourier transform is the order-zero Hankel transform (cont.)

Recall the Schläfli integral; taking $t = e^{i\theta}$ in it gives

$$J_n(x) = \frac{1}{2\pi}\int_0^{2\pi} e^{ix\sin\theta}\, e^{-in\theta}\, d\theta, \qquad \text{so for } n = 0, \quad J_0(x) = \frac{1}{2\pi}\int_0^{2\pi} e^{ix\sin\theta}\, d\theta.$$

Now suppose f is *radially symmetric*, $f(r,\theta) \equiv f(r)$. Then in (6.7.10) the θ-integral acts only on the exponential:

$$\begin{aligned} F(k_x, k_y) &= \frac{1}{2\pi}\int_0^{2\pi}\int_0^{\infty} f(r)\, e^{-ikr\sin\theta}\, r\, dr\, d\theta \\ &= \int_0^{\infty} f(r)\, r \left(\frac{1}{2\pi}\int_0^{2\pi} e^{-ikr\sin\theta}\, d\theta \right) dr = \int_0^{\infty} f(r)\, J_0(kr)\, r\, dr. \end{aligned}$$

The sign in the inner exponential does not matter over a full period. Thus

$$F(k_x, k_y) = \int_0^{\infty} f(r)\, J_0(kr)\, r\, dr \qquad (\text{no surviving } 1/2\pi).$$

The 2-D Fourier transform of a radial function is exactly its order-zero Hankel transform.

Table of contents

Useful formulas for the Bessel function

differential equation	$x^2 J_m'' + x J_m' + (x^2 - m^2)\, J_m = 0$
series solution ($m \geq 0$)	$J_m(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,(m+k)!} \left(\frac{x}{2}\right)^{2k+m}$
generating function	$e^{\frac{1}{2}x(t-t^{-1})} = \sum_{m=-\infty}^{\infty} J_m(x)\, t^m$
recursion formula	$2J_m'(x) = J_{m-1}(x) - J_{m+1}(x)$
recursion formula	$J_{m+1}(x) + J_{m-1}(x) = \frac{2m}{x}\, J_m(x) \quad (x \neq 0)$
Schläfli ($m \in \mathbb{Z}$)	$J_m(x) = \frac{1}{2\pi i} \oint e^{(x/2)(t-t^{-1})}\, t^{-(m+1)}\, dt$
integral representation	$J_m(x) = \frac{1}{\pi} \int_0^{\pi} \cos(m\theta - x \sin\theta)\, d\theta$
	$J_0(x) = \frac{1}{\pi} \int_0^{\pi} \cos(x \sin\theta)\, d\theta$

Useful formulas for the Bessel function (cont.)

Asymptotic behavior:

$$J_m(x) \approx \frac{1}{\Gamma(m+1)} \left(\tfrac{x}{2}\right)^m \qquad (x \to 0^+,\ \ m \geq 0),$$

$$J_m(x) \approx \sqrt{\frac{2}{\pi x}} \cos\!\left(x - \frac{m\pi}{2} - \frac{\pi}{4}\right) \qquad (x \to +\infty),$$

$$Y_m(x) \approx \sqrt{\frac{2}{\pi x}} \sin\!\left(x - \frac{m\pi}{2} - \frac{\pi}{4}\right) \qquad (x \to +\infty).$$

Orthogonality on the disk ($\alpha_{m,j}$ the positive zeros of J_m):

$$\int_0^1 J_m(\alpha_{m,j}x)\, J_m(\alpha_{m,k}x)\, x\, dx = \tfrac{1}{2} J_{m+1}^2(\alpha_{m,j})\, \delta_{jk}.$$

Expansion coefficient:

$$a_j = \frac{2}{J_{m+1}^2(\alpha_{m,j})} \int_0^1 f(r)\, J_m(\alpha_{m,j} r)\, r\, dr.$$